

Credit Score Classification using Python

Aim

To build a **Credit Score Classification** model using Python that classifies customers into different credit score categories based on their financial information.

Apparatus Required

- Google Colab / Jupyter Notebook
- Python 3
- Libraries:
 - Pandas
 - NumPy
 - Scikit-learn (sklearn)

Theory

Credit Score Classification is a **supervised machine learning** task used to predict the creditworthiness of customers. Based on customer attributes such as age, income, loan amount, and previous payment history, the model classifies customers into categories like **Good**, **Standard**, or **Poor** credit score.

For this experiment, a **Decision Tree Classifier** is used because it is simple, interpretable, and effective for classification problems.

Algorithm

1. Import the required libraries.
2. Create a sample credit score dataset.
3. Split the dataset into training and testing sets.
4. Train the Decision Tree Classifier.
5. Predict the credit score class for the test data.
6. Calculate the Accuracy.

7. Display the Confusion Matrix and Classification Report.

Code (Google Colab)

```
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Sample dataset
data = {
    'Age': [25, 35, 45, 20, 40, 30, 50, 23, 60, 48],
    'Income': [30000, 60000, 80000, 20000, 75000, 50000, 90000, 25000, 100000, 85000],
    'Loan': [10000, 20000, 15000, 5000, 18000, 12000, 25000, 7000, 30000, 20000],
    'CreditScore': ['Poor', 'Good', 'Good', 'Poor', 'Good',
                    'Standard', 'Good', 'Poor', 'Good', 'Standard']
}

df = pd.DataFrame(data)

X = df[['Age', 'Income', 'Loan']]
y = df['CreditScore']

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
```

```
# Train model
model = DecisionTreeClassifier(random_state=42)
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Results
print("Predicted Credit Score:")
print(y_pred)

print("\nAccuracy:")
print(accuracy_score(y_test, y_pred))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

Sample Output

```

... Predicted Credit Score:
['Good' 'Standard' 'Poor']

Accuracy:
0.3333333333333333

Confusion Matrix:
[[1 0 1]
 [0 0 0]
 [0 1 0]]

Classification Report:

```

	precision	recall	f1-score	support
Good	1.00	0.50	0.67	2
Poor	0.00	0.00	0.00	0
Standard	0.00	0.00	0.00	1
accuracy			0.33	3
macro avg	0.33	0.17	0.22	3
weighted avg	0.67	0.33	0.44	3

Result

The **Credit Score Classification** model was successfully implemented in Python using a **Decision Tree Classifier**. The model classified customers into **Good, Standard, and Poor** credit score categories and its performance was evaluated using **Accuracy, Confusion Matrix, and the Classification Report**.